

TONGULAH, Russian Federation

WMO: 248430

Lat: 61.917N

Lon: 124.550E

Elev: 199

StdP: 98.96

Time Zone: 9.00 (E09)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -51.9	(c) -49.4	(d) -53.5	(e) 0.0	(f) -52.0	(g) -51.0	(h) 0.0	(i) -49.4	(j) 4.5	(k) -28.0	(l) 3.9	(m) -28.2	(n) 0.0	(o) 20	(p) 0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	29.8	19.1	27.4	18.0	25.1	16.6	20.6	27.5	19.0	25.8	17.5	23.4	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.4	23.8	16.4	12.0	21.8	15.0	10.9	20.7	60.5	27.8	55.1	26.1	50.3	23.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.3	(b) 4.5	(c) 3.9	(e) -53.3	(f) 32.7	(g) 3.0	(h) 1.6	(i) -55.5	(j) 33.8	(k) -57.2	(l) 34.7	(m) -58.9	(n) 35.6	(o) -61.1	(p) 36.8		
(a) 5.3	(b) 4.5	(c) 3.9	(e) -53.3	(f) 23.1	(g) 3.0	(h) 2.3	(i) -55.5	(j) 24.8	(k) -57.2	(l) 26.1	(m) -58.9	(n) 27.4	(o) -61.1	(p) 29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.8	-36.3	-30.5	-18.4	-6.5	5.4	13.8	15.8	12.5	3.9	-7.8	-25.4	-33.7	(6)
(7)		DBStd	19.76	8.30	7.39	8.10	6.87	4.76	4.18	4.19	3.82	4.45	6.69	8.37	8.84	(7)
(8)		HDD10.0	7295	1437	1135	881	494	156	12	4	18	188	551	1062	1356	(8)
(9)		HDD18.3	9945	1695	1368	1139	744	401	144	101	183	433	809	1312	1614	(9)
(10)		CDD10.0	426	0	0	0	0	13	126	185	97	5	0	0	0	(10)
(11)		CDD18.3	35	0	0	0	0	0	8	23	3	0	0	0	0	(11)
(12)		CDH23.3	881	0	0	0	0	9	254	483	130	3	0	0	0	(12)
(13)		CDH26.7	267	0	0	0	0	2	68	167	30	0	0	0	0	(13)
(14)	Wind	WSAvg	1.2	0.4	0.8	1.3	1.7	1.8	1.5	1.3	1.2	1.3	1.3	1.0	0.6	(14)
(15)	Precipitation	PrecAvg	399	15	12	12	18	39	49	65	63	49	33	25	18	(15)
(16)		PrecMax	544	32	30	27	40	65	90	150	164	90	60	54	37	(16)
(17)		PrecMin	300	7	2	3	9	13	18	19	15	13	12	8	5	(17)
(18)		PrecStd	58	6	6	5	7	14	18	26	28	17	10	10	6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.9	-7.7	6.2	13.2	25.0	31.1	33.1	30.0	23.0	8.6	0.4	-6.5	(19)
(20)			MCWB	-12.4	-8.7	1.6	6.2	14.0	18.3	20.0	19.9	13.6	4.3	-1.1	-7.3	(20)
(21)		2%	DB	-16.3	-11.4	1.5	9.5	21.2	28.7	30.8	26.5	18.6	5.2	-4.1	-10.9	(21)
(22)			MCWB	-16.7	-12.0	-1.7	3.9	11.6	17.5	20.4	17.8	11.7	3.1	-4.9	-11.4	(22)
(23)		5%	DB	-19.4	-14.6	-1.5	7.1	18.6	26.3	28.5	24.1	15.7	3.1	-7.9	-16.0	(23)
(24)			MCWB	-19.8	-15.2	-3.8	2.0	10.1	16.2	19.5	16.3	10.2	0.9	-8.6	-16.4	(24)
(25)		10%	DB	-23.2	-18.1	-4.1	5.0	15.3	24.1	25.9	21.7	12.9	1.4	-11.8	-20.4	(25)
(26)			MCWB	-23.4	-18.5	-5.9	1.0	8.4	15.5	18.2	15.2	8.3	-0.2	-12.3	-20.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.4	-8.7	2.1	6.4	14.2	21.4	23.7	20.9	14.4	5.1	-1.2	-7.3	(27)
(28)			MCDWB	-11.9	-7.8	6.1	12.8	24.6	26.5	28.9	27.6	20.3	7.1	0.3	-6.4	(28)
(29)		2%	WB	-16.7	-12.0	-1.2	4.4	12.0	18.9	21.3	18.8	12.7	3.3	-4.9	-11.5	(29)
(30)			MCDWB	-16.3	-11.5	1.4	9.1	19.8	26.5	27.9	24.8	17.5	5.4	-4.1	-11.0	(30)
(31)		5%	WB	-19.7	-15.2	-3.8	2.5	10.6	17.1	20.0	17.3	10.9	1.2	-8.6	-16.5	(31)
(32)			MCDWB	-19.3	-14.6	-1.6	6.3	17.6	24.8	27.3	22.5	15.3	2.7	-7.9	-16.1	(32)
(33)		10%	WB	-23.4	-18.5	-5.9	1.0	8.9	15.8	18.6	15.9	8.9	-0.2	-12.3	-20.7	(33)
(34)			MCDWB	-23.2	-18.1	-4.1	4.6	14.9	22.5	25.0	20.9	12.1	1.3	-11.8	-20.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	14.6	18.3	16.1	12.7	14.8	15.4	13.6	9.6	10.8	9.3	(35)	
(36)			MCDBR	12.8	15.5	17.0	17.2	18.5	19.6	19.8	20.5	19.8	9.9	11.9	12.1	(36)
(37)			MCWBR	12.5	14.9	14.0	12.2	10.4	9.8	10.6	12.1	13.4	7.3	11.4	12.0	(37)
(38)		5% WB	MCDBR	12.9	15.5	16.8	14.7	16.9	17.3	17.4	17.1	18.2	8.8	11.9	12.5	(38)
(39)			MCWBR	12.5	14.8	14.1	10.5	10.0	9.4	10.4	10.5	12.8	7.0	11.3	12.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.222	0.262	0.296	0.345	0.398	0.414	0.549	0.503	0.352	0.284	0.222	0.192	(40)	
(41)		taud	1.957	2.064	2.085	2.101	2.215	2.267	1.854	1.933	2.361	2.337	1.993	1.879	(41)	
(42)		Ebn at Noon	449	691	803	833	817	810	669	662	748	680	441	290	(42)	
(43)		Edh at Noon	39	73	103	126	124	120	178	151	81	56	36	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.36	1.38	3.13	5.14	5.38	5.89	5.32	4.01	2.47	1.22	0.53	0.17	(44)	
(45)		RadStd	0.02	0.06	0.10	0.18	0.39	0.41	0.32	0.27	0.28	0.12	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.80	N/A	N/A	N/A	N/A	N/A	-271	-278	N/A	N/A

Nomenclature: See separate page